EXAM PAPERS PRACTICE

Mark schemes

Q1.

(a) $(F =) 9\mathbf{i} - 3\mathbf{j} + 5\mathbf{i} + 8\mathbf{j} - 7\mathbf{i} + 3\mathbf{j} = 7\mathbf{i} + 8\mathbf{j}$

M1: Adding the three forces with one component correct.
A1: Correct answer.

M1A1

2

(b) $(F =) \sqrt{7^2 + 8^2} = \sqrt{113} = 10.6$ N

M1: Finding magnitude with a + sign.

A1F: Correct magnitude. Accept AWRT 10.63 and $\sqrt{113}$
Follow through incorrect answers to part (a).

M1A1F

2

(c) $(a =) \dfrac{\sqrt{113}}{5} = 2.13$ m s^{-2}

M1: Dividing their force from part (a) or magnitude by 5.
A1F: Correct acceleration.

Accept 2.12 (from truncation or 10.6 / 5) or $\dfrac{\sqrt{113}}{5}$ *or AWRT 2.13.*
Follow through incorrect answers to parts (a) and (b).
Seeing just $\mathbf{a} = 1.4\mathbf{i} + 1.6\mathbf{j}$ *scores M1 A0*

M1A1F

2

(d) $\cos\alpha = \dfrac{7}{\sqrt{113}}$ or $\dfrac{7}{10.6}$

M1: Trig equation to find the angle with: cos with 7 or 8 in the numerator and $\sqrt{113}$ *in denominator*

OR

$\sin\alpha = \dfrac{8}{\sqrt{113}}$ or $\dfrac{8}{10.6}$

sin with 7 or 8 in the numerator and $\sqrt{113}$ *in denominator*

OR

$\tan\alpha = \dfrac{8}{7}$

tan with 7 and 8 in any position

A1F: Correct equation.

M1A1F

(α =)48.8°

A1F: Correct angle. Accept 49° or AWRT 49°
Follow through incorrect answers to parts (a) and (b).

A1F

3

[9]

Q2.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\mathbf{r}$.

A1: Correct expression for $\mathbf{r}$*. Allow equivalent column vector answer.*

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

M1: Putting their $\mathbf{j}$ *component equal to zero to form a quadratic equation.*
A1: Correct equation.

M1A1

$t = 30$ seconds

A1: For 30 seconds. No need to see $t = 0$.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t.
Can be implied by subsequent working in terms of t.

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For 0.1t – 1 = ±(3 – 0.2t). May be with their

components if velocity stated incorrectly.
A1: Correct equation.

M1A1

$t = 20$

A1: $t = 20$

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases
If the equation in t in line 2 is not seen:
then seeing $t = 20$ and $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j}$ and $v = 1.41$
award 4 out of 6
or
then seeing $t = 20$ and $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j}$ award 2 out of 6

6

[11]

Q3.

(a) $\mathbf{v} = (0.5\mathbf{i} + 0.375\mathbf{j}) \times 20 \ (= 10\mathbf{i} + 7.5\mathbf{j})$

M1: Calculating velocity with $\boldsymbol{u} = 0\boldsymbol{i} + 0\boldsymbol{j}$ and $t = 20$.
A1: Correct expression for velocity.

M1A1

$$v = \sqrt{10^2 + 7.5^2} = 12.5 \text{ ms}^{-1}$$

dM1: Calculating speed.
A1: Correct speed.

dM1A1

4

(b) $$\tan\theta = \frac{0.5}{0.375} \text{ or } \frac{10}{7.5} \left(\text{or } \tan\theta = \frac{0.375}{0.5} \text{ or } \frac{7.5}{10} \right)$$

M1: Using trig to find angle. Can be inverted as shown in brackets.
A1F: Correct trig expression with any correct equivalent fraction.

M1A1F

$\theta = 053°$

A1: Correct angle to the nearest degree. Accept 53°.

A1

OR

$$\cos\theta = \frac{7.5}{12.5} \text{ or } \frac{0.375}{0.625} \left(\text{or } \cos\theta = \frac{10}{12.5} \right)$$

(M1A1F)

θ = 053°

Note: For 37° award M1A0A0
But for 90 – 37 = 53° award M1A1A1.
For 127°, award M1A1A0

(A1)

OR

$$\sin\theta = \frac{10}{12.5} \text{ or } \frac{0.5}{0.625} \left(\text{or } \sin\theta = \frac{7.5}{12.5} \right)$$

Note: 53.1° as final answer scores M1A1A0

(M1A1F)

θ = 053°

Condone finding angle from acceleration or position vector.

(A1)

3

(c) ($\mathbf{r}$ =) $\frac{1}{2}$ (0.5$\mathbf{i}$ + 0.375$\mathbf{j}$) t^2 (= 0.25 t^2 $\mathbf{i}$ + 0.1875t^2 $\mathbf{j}$)

M1: Finding an expression for position vector in terms of t.
A1: Correct position vector.

M1A1

$500^2 = (0.25\, t^2)^2 + (0.1875\, t^2)^2$

dM1: Using distance to form an equation for t.
A1: Correct equation.

dM1A1

$$t = 4\sqrt{\frac{500^2}{0.25^2 + 0.1875^2}} = 40 \text{ seconds}$$

A1: Correct time.

A1

OR

$a = 0.625$

M1: Finding magnitude of acceleration.
A1: Correct acceleration

(M1A1)

$$500 = \frac{1}{2}0.625t^2$$

dM1: Using distance to form an equation for t.
A1: Correct equation.

(dM1A1)

$t = 40$

A1: Correct time.

(A1)

OR

$400 = \frac{1}{2}\ 0.5t^2$ *or* $300 = \frac{1}{2}\ 0.375t^2$

M1: Working with one component.
A1: Correct distance (300 or 400)
A1: Correct equation.

(M1A1)
(A1)

$t^2 = 1600$

dM1: Solving for t.

(dM1)

$t = 40$

A1: Correct t.
Note: 500 ÷ 12.5 = 40 is not acceptable and scores 0

(A1)

5

[12]

Q4.

(a) **F** = 5**j** + 8**i** – 7**j** = 8**i** – 2**j**

Adding the two forces. For incorrect answers, evidence of adding must be seen

M1

Correct resultant

A1

2

(b) $F = \sqrt{8^2 + 2^2} = \sqrt{68} = 8.25\ \text{N}$

Finding magnitude (must see addition and not subtraction)

M1

Correct magnitude

Accept $2\sqrt{17}, \sqrt{68}$ *or AWRT 8.25 (eg 8.246)*

A1F

2

(c)

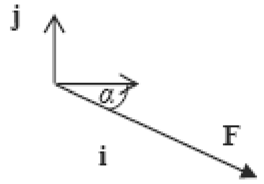

Diagram with force in the correct quadrant and with correct direction shown by an arrow.

B1

$\tan \alpha = \frac{2}{8}$

Using trig to find angle: if tan, 8 in denominator; if sin or cos, 8.25 or their answer to part (b) in denominator

M1

$\alpha = 14.0°$

Correct angle
Accept 14.1 or 14 or AWRT 14.0 (eg 14.04)
M1 and A1 not dependent on B1

A1

3

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Q5.

(a) **u** = 5**i** or $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$

Correct velocity

B1

1

(b) **v** = 5**i** + (–0.2**i** + 0.25**j**)t

*Use of constant acceleration equation, with **u** and **a** not zero*

M1

Correct velocity
*M1A0 for using 5**j** or just 5*

A1

OR

$\mathbf{v} = \begin{bmatrix} 5 - 0.2t \\ 0.25t \end{bmatrix}$

2

(c) $5 - 0.2t = 0$

Easterly component zero

M1

Correct equation

A1

$t = \frac{5}{0.2} =$ 25 seconds

Correct t

A1

3

(d) $\mathbf{r} = 5\mathbf{i} \times 25 + \frac{1}{2}(-0.2\mathbf{i} + 0.25\mathbf{j}) \times 25^2$

Use of constant acceleration equation with t from part (c)

M1

Correct expression based on t from part (c)

A1F

$= 62.5\mathbf{i} + 78.125\mathbf{j}$

Correct simplification CAO

A1

$\theta = \tan^{-1}\left(\frac{62.5}{78.125}\right)$

Using tan to find the angle

dM1

Correct expression based on t from part (c), *with correct two values (either way)*

A1F

= 038.7°

Correct angle
Accept 38.6° or 039°

A1

OR

$\mathbf{r} = \frac{1}{2}(5\mathbf{i} + 6.25\mathbf{j}) \times 25$

(M1)(A1F)(A1)

$$\theta = \tan^{-1}\left(\frac{5}{6.25}\right) = 038.7°$$

(dM1)(A1F)(A1)

6

[12]

Q6.

(a) $\mathbf{v} = 4\mathbf{i} + (-3\mathbf{i} + 12\mathbf{j})t$

M1

use of $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{a}t$

A1

2

(b) $t = 0.5$, $\mathbf{v} = 2.5\mathbf{i} + 6\mathbf{j}$

Ft 2 terms and t subs

B1ft

Speed = $\sqrt{(2.5^2 + 6^2}$

2 terms

M1

Speed = 6.5 ms^{1}

ft 2 terms

A1ft

3

[5]

Q7.

(a) $\mathbf{F} = \begin{pmatrix} 6 \\ -2.5 \end{pmatrix}$ N

B1 each component

B1 B1

2

(b) $|\mathbf{F}| = \sqrt{6^2 + (-2.5)^2}$

Must see +

M1

= 6.5 N

ft from vector in (a)

A1F

2

[4]

Q8.

(a) (i) vertical: $s = ut + \frac{1}{2}at^2$

$0 = 7t - 4.9t^2$

Full method

M1

$0 = t(7 - 4.9t)$

$0 = 7 - 4.9t$

m1

$t = \frac{10}{7}$ sec (1.43)

A1

3

(ii) $OF = 21 \times \frac{10}{7}$

M1

= 30 m

A1F

2

(b) vert : $0 = 3.5t - 4.9t^2$

Full method required

M1

$t = \frac{3.5}{4.9}$

m1

$FG = 21 \times \frac{3.5}{4.9}$

M1

= 15 m

A1

4

(c) $GH = \frac{1}{2} \times 15 = 7.5$

B1F

$OH = 30 + 15 + 7.5$

M1

= 52.5 m

A1F

3

[12]

Q9.

(a) (i) $\mathbf{a} = \frac{1}{4}(8\mathbf{i} - 3\mathbf{j}) = 2\mathbf{i} - 3\mathbf{j}$

Use of $\mathbf{F} = m\mathbf{a}$*, must be applied to both components*

M1

Correct acceleration

A1

2

(ii) $\mathbf{v} = 20(2\mathbf{i} - 3\mathbf{j}) = 40\mathbf{i} - 60\mathbf{j}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ *with* $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

*Correct **v***

A1

2

(iii) $\mathbf{r} = \frac{1}{2}(2\mathbf{i} - 3\mathbf{j}) \times 20^2$

*Use of constant acceleration equation to find **r** with **u** = 0**i** + 0**j***

M1

Correct expression

A1

$= 400\mathbf{i} - 600\mathbf{j}$

*Correct **r** in simplified form*

A1

3

(b) $\mathbf{r} = 400\mathbf{i} - 600\mathbf{j} + 25(40\mathbf{i} - 60\mathbf{j})$

*Use of **r** + 25**v***

M1

Correct expression

A1ft

$= 1400\mathbf{i} - 2100\mathbf{j}$

Correct position vector

A1ft

$r = \sqrt{1400^2 + 2100^2}$ = 2520 m (to 3 sf)

Finding magnitude

m1

Correct distance from correct working

A1ft

5

***Alternative:** Straight line method*
M1 two distances r and 25v
A1 for each distance
m1 adding
A1 correct final answer

[12]

Q10.

(a) $\mathbf{F} = (3\mathbf{i} + 4\mathbf{j}) + (6\mathbf{i} - 8\mathbf{j})$

Addition of the two forces

M1

2

$= 9\mathbf{i} - 4\mathbf{j}$

Correct resultant

A1

(b) $F = \sqrt{9^2 + 4^2} = 9.85 \text{ N}$

Finding magnitude

M1

Correct magnitude

A1

2

(c) $\tan \alpha = \frac{4}{9}$

Using tan to find the angle

M1

Correct equation

A1

$\alpha = 24.0°$

Correct angle

A1

3

[7]